

PROJECT PROPOSAL: Next-Generation Quantum Key Distribution (QKD) Protocols

Abstract:

This project investigates post-quantum cryptographic primitives combined with high-efficiency satellite-to-ground quantum communications. Our objective is to demonstrate robust entanglement generation over turbulent atmospheric channels.

Novelty:

Existing QKD deployments rely on fiber links limited to ~100 km. We introduce a novel adaptive optics phase correction scheme operating at 1550 nm.

Technical Feasibility & Architecture:

- Subsystem A: Entangled Photon Pair Source (SPDC crystal, 80 MHz repetition rate)
- Subsystem B: Fast Steering Mirrors with closed-loop piezo actuation
- Subsystem C: Superconducting Nanowire Single-Photon Detectors (SNSPDs)

Risk Mitigation:

Atmospheric turbulence models will be validated via horizontal ground-test link (12 km).

Budget & Milestones:

- Phase 1 (Months 1-6): Optical bench prototype (\$180,000)
- Phase 2 (Months 7-14): Field testing and channel characterization (\$250,000)
- Phase 3 (Months 15-24): Synthesis of final evaluation report (\$70,000)

Total Grant Request: \$500,000